

Cloud Computing Practical Examination

Total Marks: 40

Duration: 60 Minutes

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Div: A

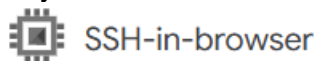
Question 1: Docker Installation & Nginx Deployment

Task

1. Install Docker on your cloud virtual machine.

```
satyamrevgade2@instance-20260609-172904:~$ ls
satyamrevgade2@instance-20260609-172904:~$ ls -a
. .bash_logout .bashrc .profile .ssh
satyamrevgade2@instance-20260609-172904:~$ mkdir test3
satyamrevgade2@instance-20260609-172904:~$ ls
test3
satyamrevgade2@instance-20260609-172904:~$ cd test3/
satyamrevgade2@instance-20260609-172904:~/test3$ cd ..
satyamrevgade2@instance-20260609-172904:~$ sudo apt-get update
Get:1 file:/etc/apt/mirrors/debian.list Mirrorlist [30 B]
Get:5 file:/etc/apt/mirrors/debian-security.list Mirrorlist [39 B]
Hit:7 https://packages.cloud.google.com/apt/google-compute-engine-bookworm-stable InRelease
Hit:2 https://deb.debian.org/debian bookworm InRelease
Get:3 https://deb.debian.org/debian bookworm-updates InRelease [55.4 kB]
Get:4 https://deb.debian.org/debian bookworm-backports InRelease [59.4 kB]
Hit:8 https://packages.cloud.google.com/apt/cloud-sdk-bookworm InRelease
Get:6 https://deb.debian.org/debian-security bookworm-security InRelease [40.0 kB]
Get:9 https://deb.debian.org/debian bookworm-backports/main Sources.diff/Index [63.3 kB]
Get:10 https://deb.debian.org/debian bookworm-backports/main amd64 Packages.diff/Index [63.3 kB]
Get:11 https://deb.debian.org/debian bookworm-backports/main Translation-en.diff/Index [63.3 kB]
Get:15 https://deb.debian.org/debian bookworm-backports/main Sources T-2026-06-12-2001.06-F-2026-06-12-2001.06.pdiff [561 B]
Get:16 https://deb.debian.org/debian bookworm-backports/main amd64 Packages T-2026-06-12-2001.06-F-2026-06-12-2001.06.pdiff [512 B]
Hit:17 https://packages.cloud.google.com/apt/google-cloud-ops-agent-bookworm-2 InRelease
Get:15 https://deb.debian.org/debian bookworm-backports/main Sources T-2026-06-12-2001.06-F-2026-06-12-2001.06.pdiff [561 B]
Get:16 https://deb.debian.org/debian bookworm-backports/main amd64 Packages T-2026-06-12-2001.06-F-2026-06-12-2001.06.pdiff [512 B]
Get:18 https://deb.debian.org/debian bookworm-backports/main Translation-en T-2026-06-12-2001.06-F-2026-06-12-2001.06.pdiff [348 B]
Get:18 https://deb.debian.org/debian bookworm-backports/main Translation-en T-2026-06-12-2001.06-F-2026-06-12-2001.06.pdiff [348 B]
Get:12 https://deb.debian.org/debian-security bookworm-security/main Sources [210 kB]
Get:13 https://deb.debian.org/debian-security bookworm-security/main amd64 Packages [312 kB]
Get:14 https://deb.debian.org/debian-security bookworm-security/main Translation-en [191 kB]
Fetched 1067 kB in 1s (822 kB/s)
Reading package lists... Done
satyamrevgade2@instance-20260609-172904:~$
```

- 2.
3. Verify Docker installation.



```
satyamrevgade2@instance-20260609-172904:~$ docker --version
Docker version 29.5.3, build d1c06ef
satyamrevgade2@instance-20260609-172904:~$
```

- 4.
3. Download the Nginx image.

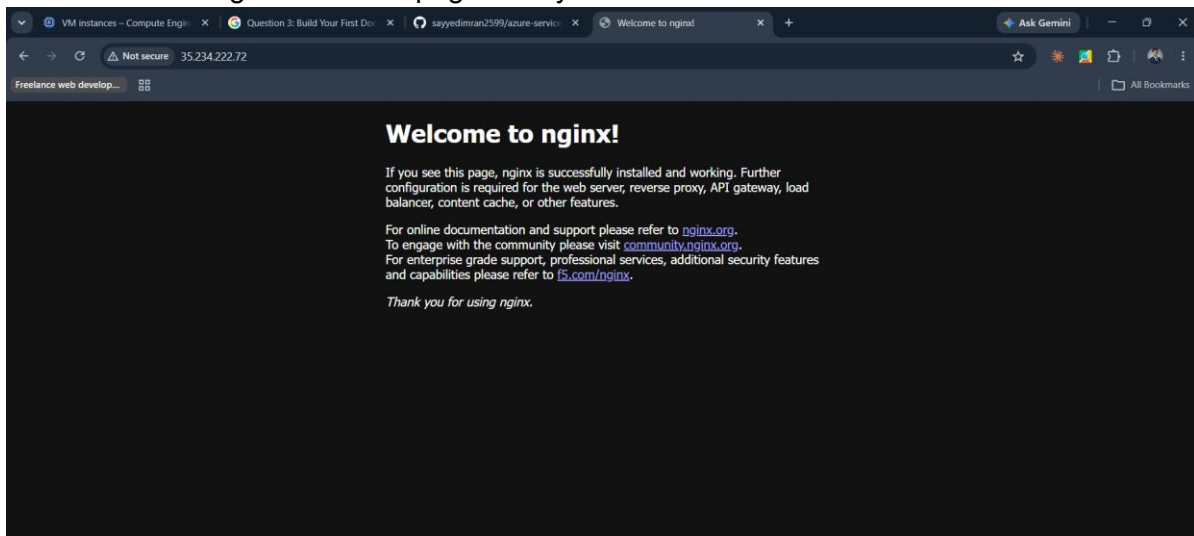
```
satyamrevgade2@instance-20260609-172904:~$ docker --version
Docker version 29.5.3, build d1c06ef
satyamrevgade2@instance-20260609-172904:~$ sudo docker pull nginx
sudo docker images
Using default tag: latest
latest: Pulling from library/nginx
d4dcde3aeeed: Pull complete
6376488be516: Pull complete
c5a7565de4cf: Pull complete
57b3f8f43092: Pull complete
80fa08b690ad: Pull complete
41c9fef90940: Pull complete
72c03230f136: Pull complete
ae32205c0ff5: Download complete
9bbdb266cee4: Download complete
Digest: sha256:608a100c71651bf5b773c89083b4alad7ef4b2bd05d7a7e552271e03123692ad
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest

IMAGE        ID                DISK USAGE    CONTENT SIZE    EXTRA
nginx:latest  608a100c7165     24MB          66MB
```

5. Run the Nginx container.

```
satyamrevgade2@instance-20260609-172904:~$ sudo docker run --name my-nginx-server -p 80:80 -d nginx
sudo docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                NAMES
bf8d8961094c  nginx    "/docker-entrypoint..." Less than a second ago    Up Less than a second    0.0.0.0:80->80/tcp, [::]:80->80/tcp    my-nginx-server
```

6. Access the Nginx welcome page from your browser.



Question 2: Docker Volume Practical

Task

1. Create a Docker volume named: studentdata

```
satyamrevgade2@instance-20260609-172904:~$ sudo docker volume create studentdata
studentdata
satyamrevgade2@instance-20260609-172904:~$
```

- 2.
3. Run an Ubuntu container using the volume.

```
satyamrevgade2@instance-20260609-172904:~$ sudo docker volume create studentdata
studentdata
satyamrevgade2@instance-20260609-172904:~$ sudo docker run --name my-ubuntu-container -v studentdata:/data -it ubuntu bash
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
1c24335ddd46: Pull complete
6f5c5aa4e145: Pull complete
9bcf140d7f0f: Download complete
Digest: sha256:f3d28607ddd78734bb7f71f117f3c6706c666b8b76cbff7c9ff6e5718d46ff64
Status: Downloaded newer image for ubuntu:latest
root@447b8a080691:/# ^C
root@447b8a080691:/# ^C
root@447b8a080691:/# ^C
root@447b8a080691:/# exit
exit
satyamrevgade2@instance-20260609-172904:~$
```

- 4.

3. Create a file: /data/info.txt
4. Write your name and roll number inside the file.
5. Display the contents of the file.

```
satyamrevgade2@instance-20260609-172904:~/azure-services$ sudo docker start my-ubuntu-container
my-ubuntu-container
satyamrevgade2@instance-20260609-172904:~/azure-services$ sudo docker exec -it my-ubuntu-container bash
root@0649f1a86b24:/# ls
bin boot data dev etc home lib lib64 media mnt opt proc root run sbin srv sys sms usr var
root@0649f1a86b24:/# cd /data
pwd
/data
root@0649f1a86b24:/data# echo "Satyam Revgade" > /data/info.txt
echo "Roll Number: YOUR_ROLL_NUMBER" >> /data/info.txt
root@0649f1a86b24:/data# cat /data/info.txt
Satyam Revgade
Roll Number: YOUR_ROLL_NUMBER
root@0649f1a86b24:/data#
```

Question 3: Build Your First Docker Image

Task

Create a custom Docker image that displays the following message when executed:

Cloud Computing Practical Exam Completed Successfully

Build the image and run it successfully.

```
satyamrevgade2@instance-20260609-172904:~/azure-services$ mkdir practicaalexam
cd practicaalexam
nano Dockerfile
satyamrevgade2@instance-20260609-172904:~/azure-services/practicaalexam$ cat Dockerfile
FROM ubuntu

CMD echo "Cloud Computing Practical Exam Completed Successfully"
satyamrevgade2@instance-20260609-172904:~/azure-services/practicaalexam$ sudo docker build -t practicaalexam .
[+] Building 0.4s (5/5) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 115B
=> WARN: JSONArgsRecommended: JSON arguments recommended for CMD to prevent unintended behavior related to OS signals (line 3)
=> [internal] load metadata for docker.io/library/ubuntu:latest
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/1] FROM docker.io/library/ubuntu:latest@sha256:f3d28607ddd78734bb7f71f117f3c6706c666b8b76cbff7c9ff6e5718d46ff64
=> => resolve docker.io/library/ubuntu:latest@sha256:f3d28607ddd78734bb7f71f117f3c6706c666b8b76cbff7c9ff6e5718d46ff64
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:f579aaa985f6f08565d2e70df24372d3c6c42237eb29b69732a42640cf8df936
=> => exporting config sha256:d5233c4ade339f2f401f7c45ba526b560b58e63e04cf8444b956d97b0e8b9e0
=> => exporting attestation manifest sha256:d50a312357a4034ff15d464a275f1fb00b85a0d95337f410b7be80231dbefc6b
=> => exporting manifest list sha256:bb6b0ca63c5056b07190037fd337e77dc81a393b92243c031e58906f1b70cdb4
=> => naming to docker.io/library/practicaalexam:latest
=> => unpacking to docker.io/library/practicaalexam:latest

1 warning found (use docker --debug to expand):
- JSONArgsRecommended: JSON arguments recommended for CMD to prevent unintended behavior related to OS signals (line 3)
```

Required Screenshots

- Dockerfile
- docker build output
- docker images output
- docker run output

Bonus Challenge (Optional)

Explain in 2–3 lines each:

1. Difference Between a Docker Image and a Docker Container

A **Docker Image** is a read-only template that contains the application code, libraries, dependencies, and configuration required to run an application. A **Docker Container** is a running instance of a Docker Image that executes the application in an isolated environment.

2. Why Are Docker Volumes Used?

Docker Volumes are used to provide persistent storage for containers. Data stored in a volume remains available even if the container is stopped, deleted, or recreated, making it useful for storing application data, logs, and configuration files.